Home lab Project Week8, Day 4

Task:
Test the complete firewall environment, verify interface assignments and IP addressing, document the topology, and correct configuration problems.

Here's how the task breaks into concrete steps.

1. Verify interface assignments

Log into the OPNsense console (`root`/`opnsense`), and at the main menu, the WAN/LAN summary should print automatically above the numbered options — confirm it shows something like:

WAN (vtnet0) -> v4: <your WAN IP>
LAN (vtnet1) -> v4: 192.168.78.1/24

Also log into the **web GUI** now (`https://192.168.78.1` from Kali's browser, logging in with `root`/`opnsense`) and go to **Interfaces** → **Overview**. This shows both interfaces, their status (should be green/up), and assigned IPs in a cleaner graphical view — better for a screenshot than the console.

2. Verify IP addressing end-to-end

Run through this checklist from **Kali**:

ip a

Confirm Kali's own IP is correctly set in the `192.168.78.x` range.

ping 192.168.78.1

Confirms Kali ↔ OPNsense LAN connectivity (you likely already proved this in Day 3, but repeat it here as part of this day's own evidence).

ping 192.168.78.5

Confirms Kali ↔ Ubuntu still works (direct, since DMZ segmentation hasn't happened yet — that's Week 10).

From OPNsense's web GUI, go to **Interfaces** → **Diagnostics** → **Ping**, and ping Kali's IP from OPNsense's side too — this proves connectivity works in both directions, not just one.

3. Document the topology

This is the deliverable piece — a simple diagram or table showing what's connected to what. Something like:

Device	Interface	VirtualBox Network	IP Address
OPNsense	WAN	NAT	(DHCP-assigned)
OPNsense	LAN	Internal Network `lab-lan`	192.168.78.20/24
Kali	eth0	Internal Network `lab-lan`	192.168.78.3/24
Ubuntu	eth0	*(still flat/unsegmented — DMZ comes in Week 10)*	192.168.78.5/24

A simple boxes-and-lines diagram works well too, even hand-drawn or in a quick draw.io/PowerPoint sketch — doesn't need to be fancy, just clear.

4. Correct configuration problems

This is where you note any issues you actually hit and fix them — and you have real material for this already:

- The Hyper-V/Virtual Machine Platform conflict that caused boot freezes, and how you resolved it
- The storage controller mix-up from the Hyper-V detour
- Anything that doesn't check out when you run the verification steps above — fix it and note what you changed

Screenshots to capture for Day 4:

- OPNsense web GUI → Interfaces → Overview (both interfaces, green/up, with IPs)
- Kali's successful pings to both OPNsense and Ubuntu
- OPNsense's Diagnostics → Ping showing it can reach Kali
- Your topology table or diagram